

Marmara University

Faculty of Engineering

Electronics 1 EE3011

Experiment (4) Diode Characteristics

Department	Student ID	Name Surname
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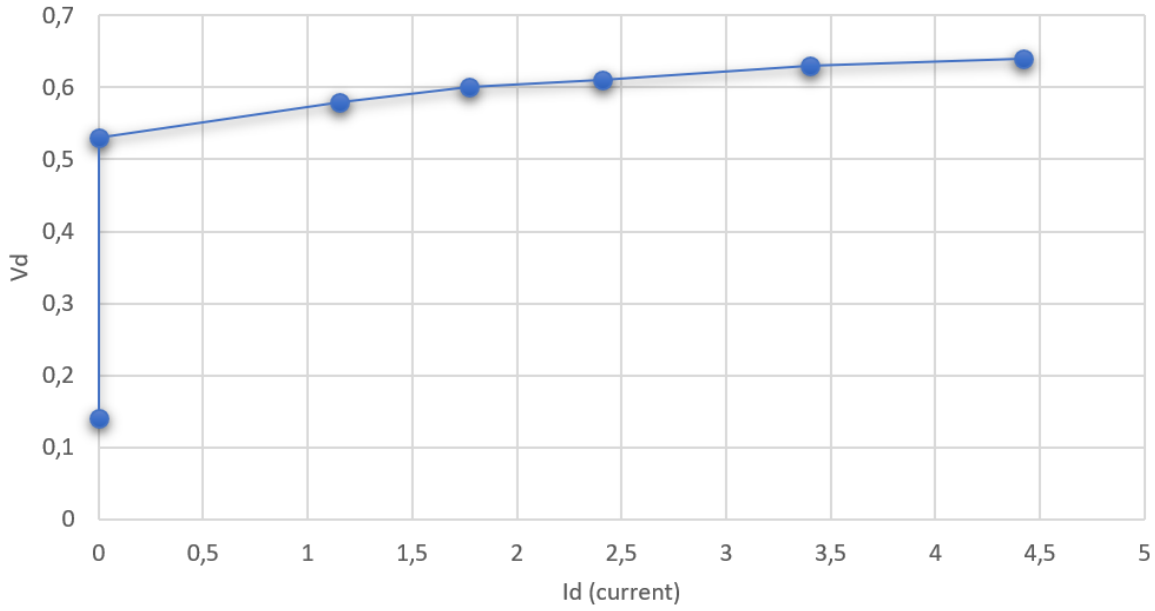
Part (1): Diode Forward IV Characteristics.

Answer:

Yes, the curve resembles the typical diode IV characteristic, starting conduction around 0.6 V. Minor discrepancies are due to measurement errors and diode tolerance.

V	Vd	VR	Id=Vr/R
0.14	0.14	0	0
0.91	0.53	0.38	0.38
1.73	0.58	1.15	1.15
2.37	0.6	1.77	1.77
3.02	0.61	2.41	2.41
4.03	0.63	3.4	3.4
5.06	0.64	4.42	4.42

Table 1: Measurements for Part (1)

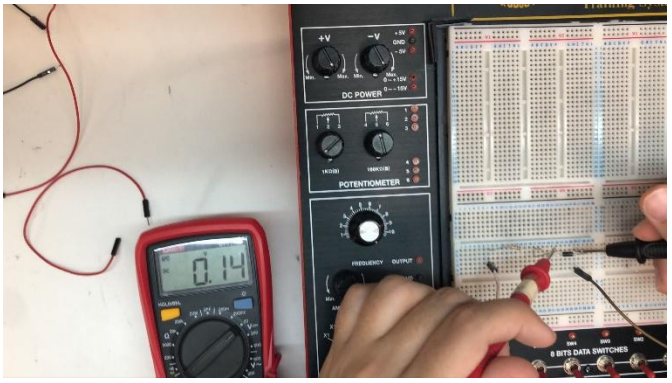


Graph 1: Diode IV Characteristics

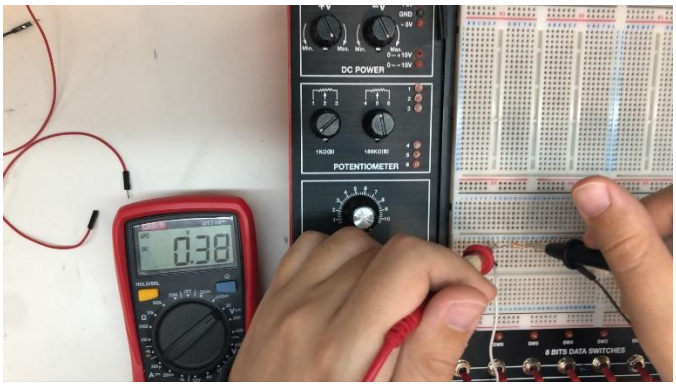
For 0.14V ⇒Vr



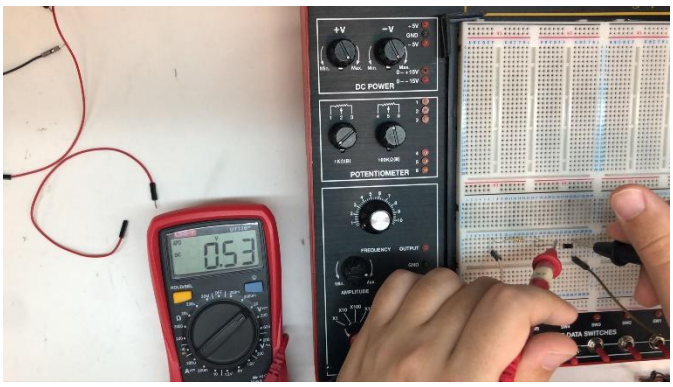
For 0.14V ⇒ Vd



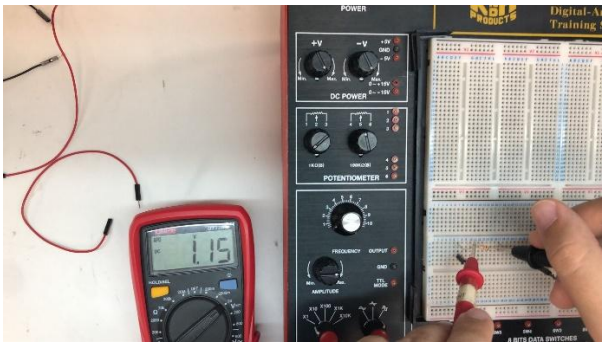
For 0.91V ⇒Vr



For 0.91V ⇒ Vd



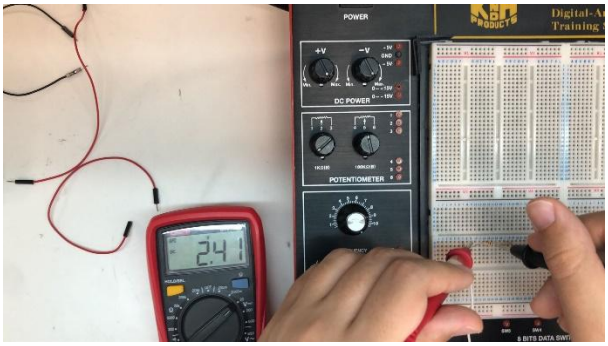
For 1.73V ⇒ Vr



For 1.73V ⇒ Vd



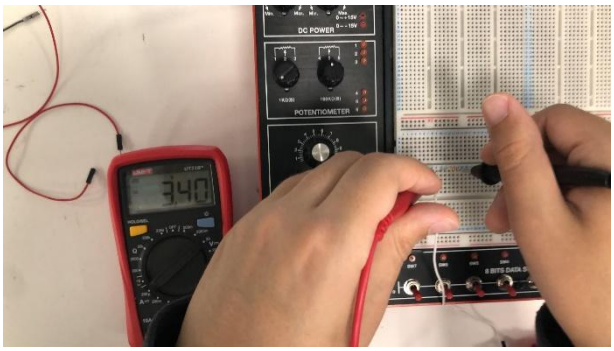
For 3.02V⇒ Vr



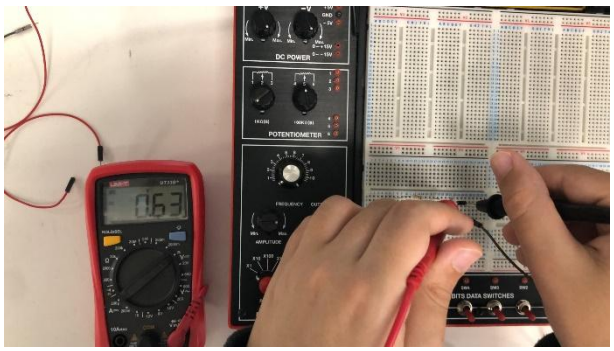
For 3.02V ⇒ Vd



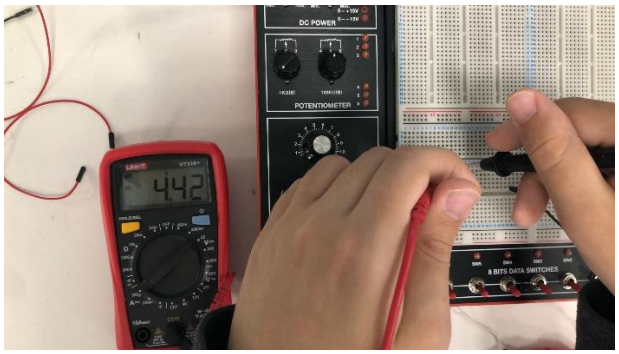
For 4.03V⇒ Vr



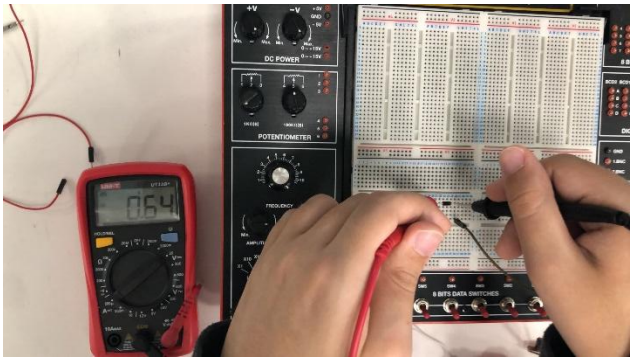
For 4.03V ⇒ Vd



For 5.06V⇒ Vr



For 5.06V ⇒ Vd



Part (2):Graphical Analysis of Diode Circuits

Diode	Vd	Vr	Id=Vr/R
Included	0.59	1.47	1.47
Removed	0	0	0
Shorted	0	2V	2

Table 2: Measurements for Part (2)

I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference	I_D (mA) Theoretical value	% Difference
1.47	1.5	2.00%	1.3	13.1%

Table 3: Calculations for Part (2)

Part (3): Diode Application as a Logical Gate

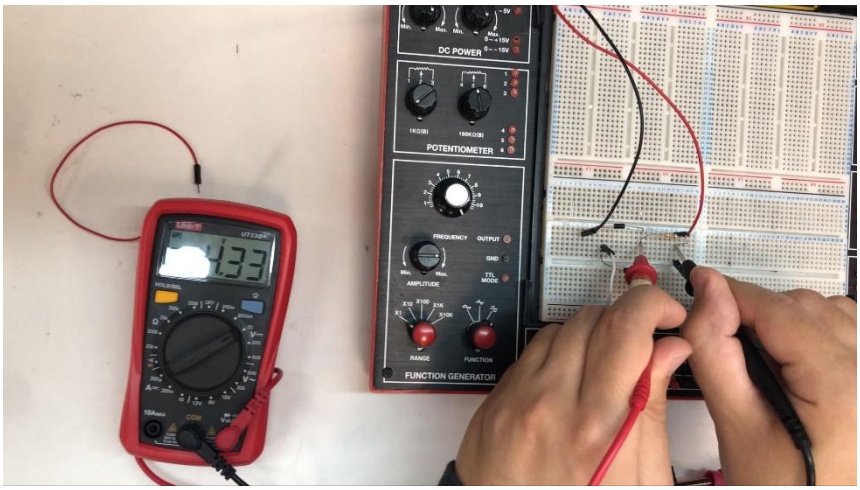
Answer:

They exhibit non-linear behavior and lack amplification capabilities. Logic gates require precise switching characteristics and amplification to ensure reliable operation, which diodes alone cannot provide. Instead, transistors, which offer both amplification and switching capabilities, are predominantly used in logic gate designs.

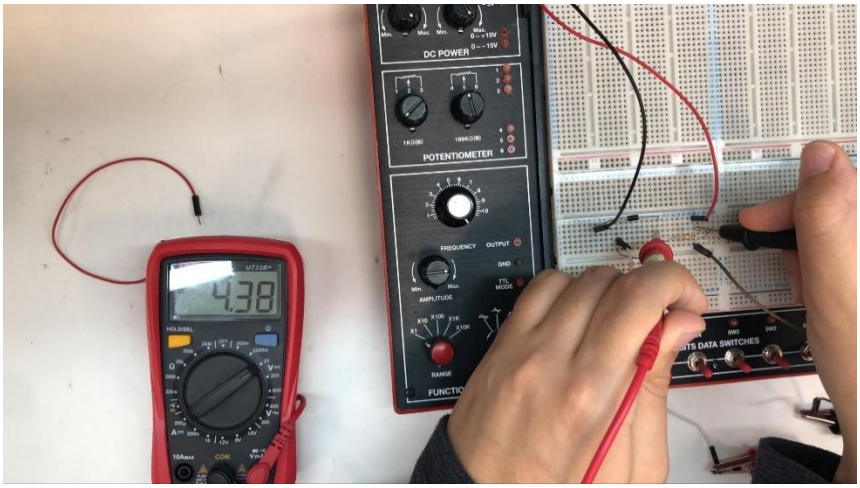
Va	Vb	Vy
0	0	0
0	5	4.33
5	0	4.38
5	5	4.4

Table 4: Measurements for Part (3).

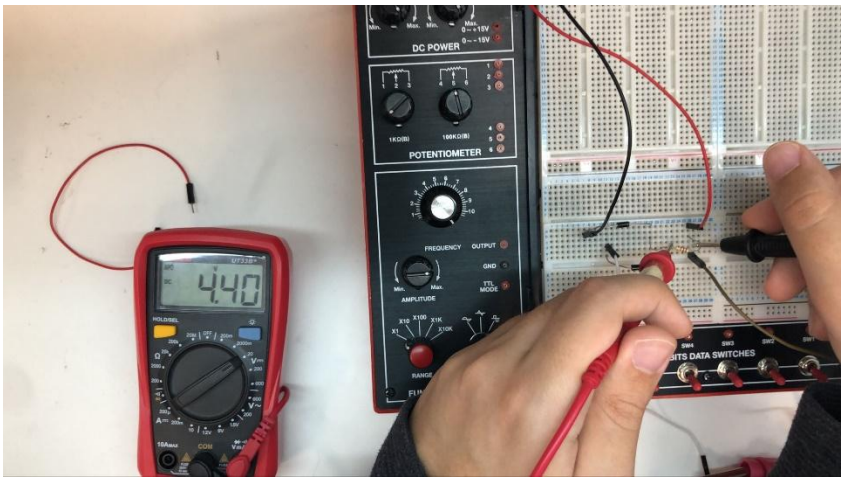
For Va=0 Vb=5 Vy=4.33



For Va=5 Vb=0 Vy=4.38

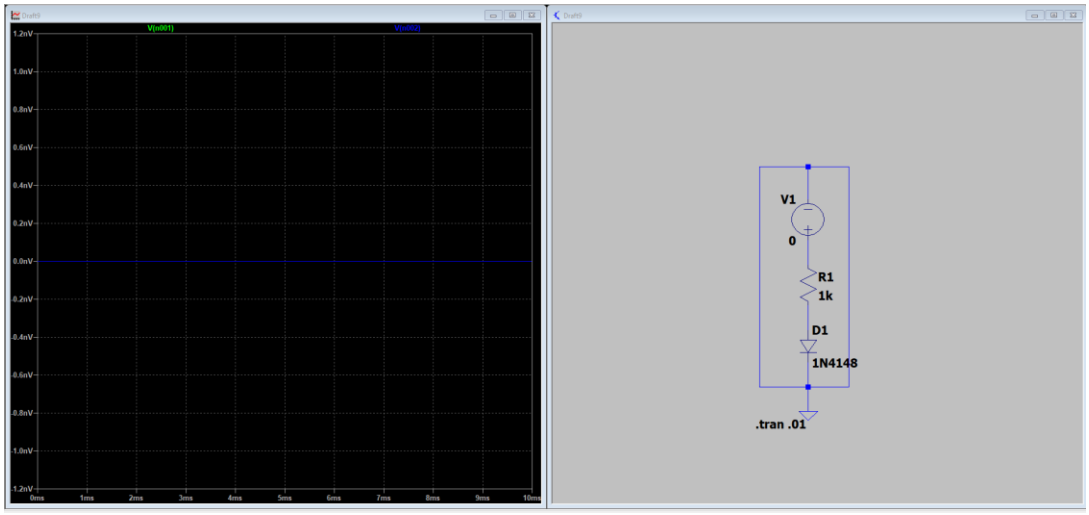


For Va=5 Vb=5 Vy=4.4

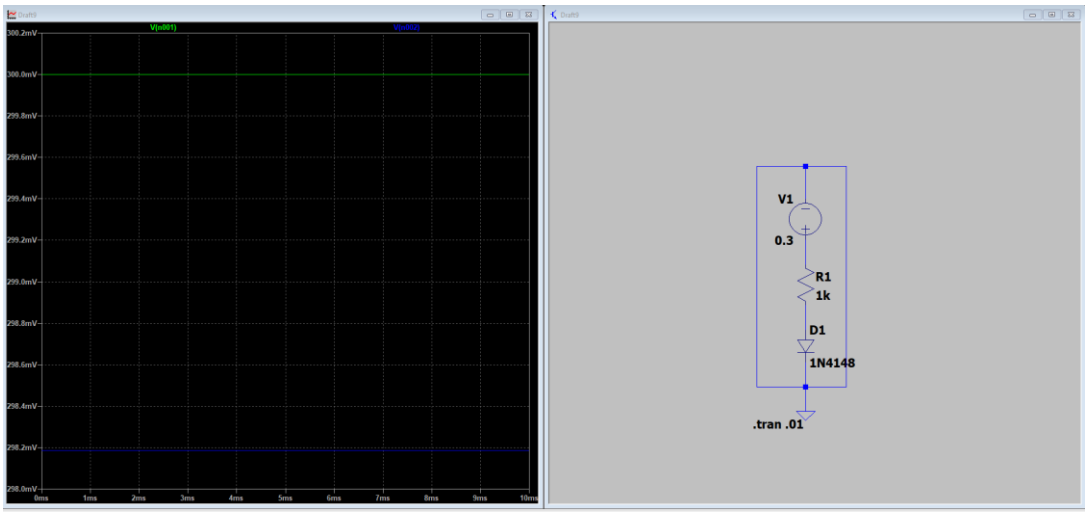


LTspice Simulations and Conclusions and Remarks

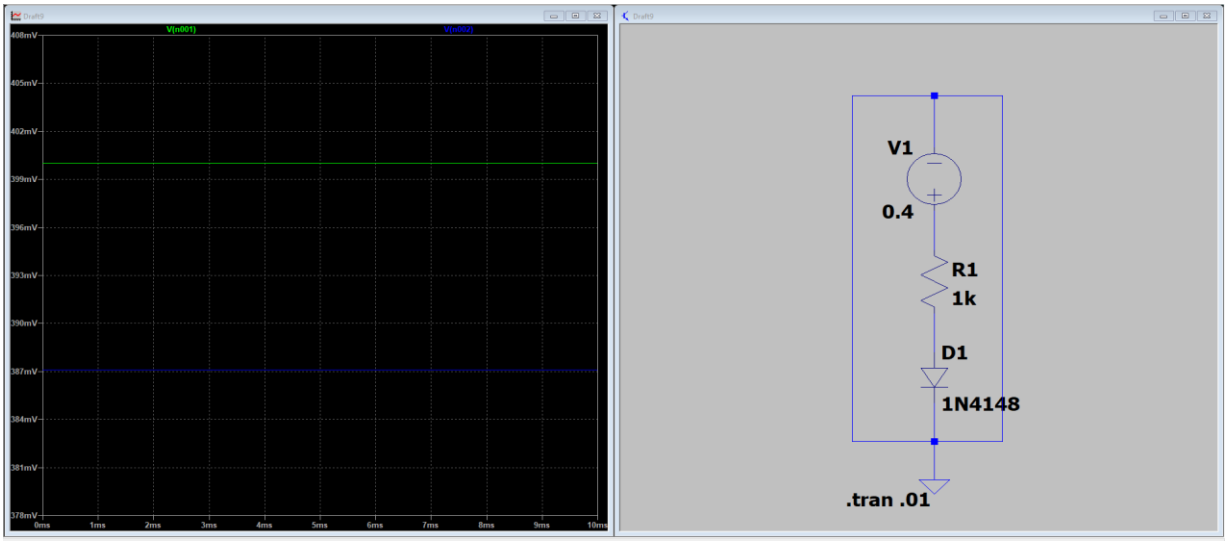
Part 1



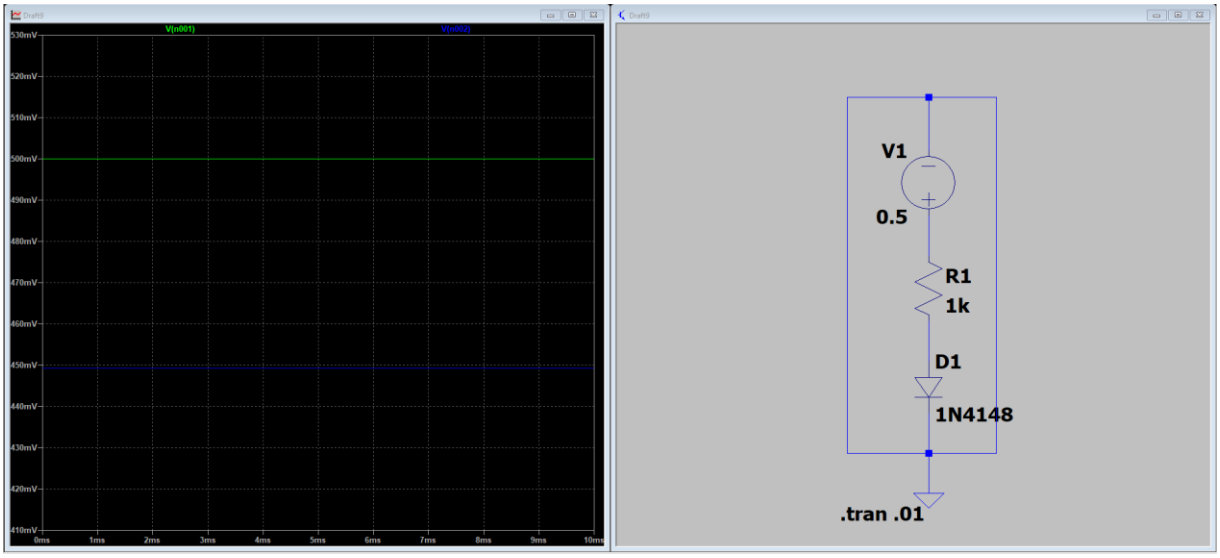
0V



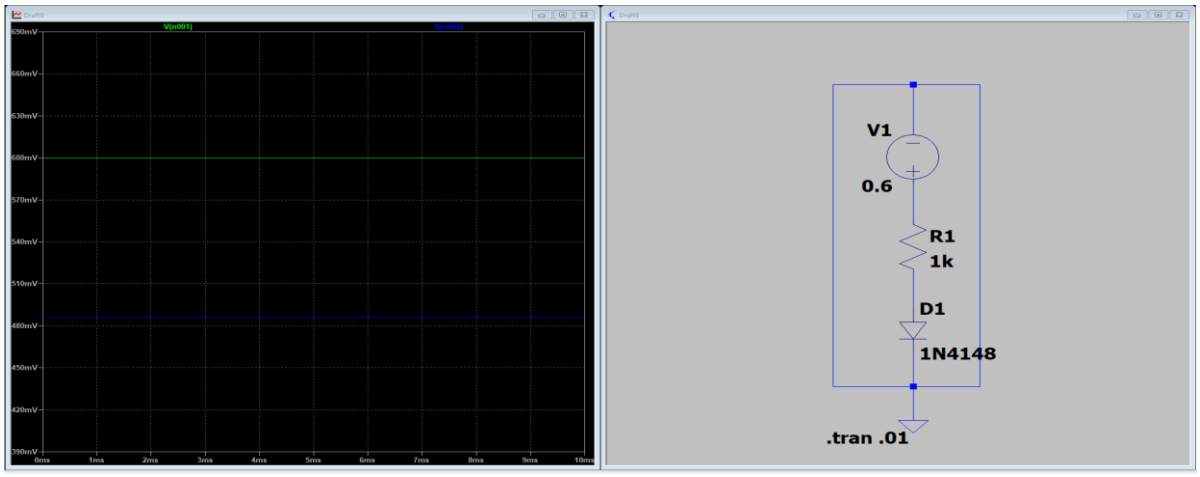
0.3V


V(n001)
V(n002)
408mV
405mV
402mV
399mV
396mV
393mV
390mV
387mV
384mV
381mV
378mV
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
V1
0.4
R1
1k
D1
1N4148
.tran .01

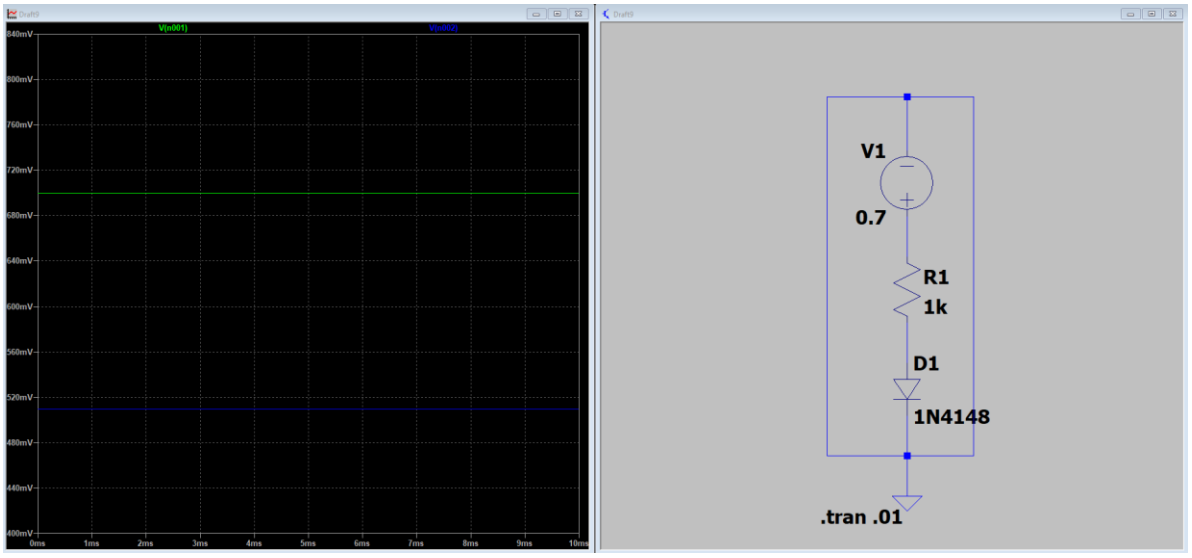
0.4 V


V(n001)
V(n002)
530mV
520mV
510mV
500mV
490mV
480mV
470mV
460mV
450mV
440mV
430mV
420mV
410mV
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
V1
0.5
R1
1k
D1
1N4148
.tran .01

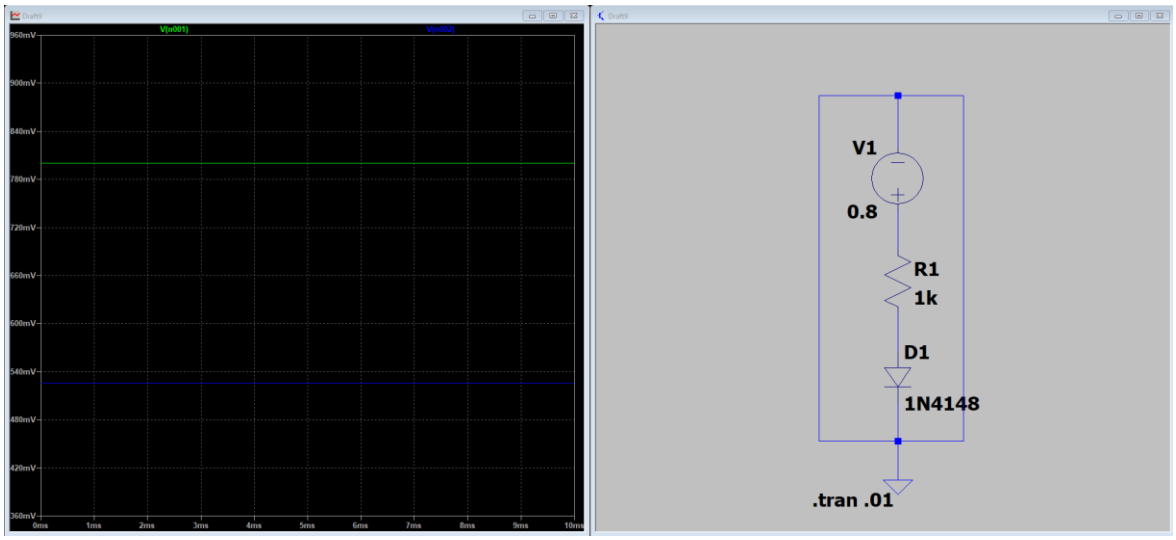
0.5V


V1
0.6
R1
1k
D1
1N4148
.tran .01

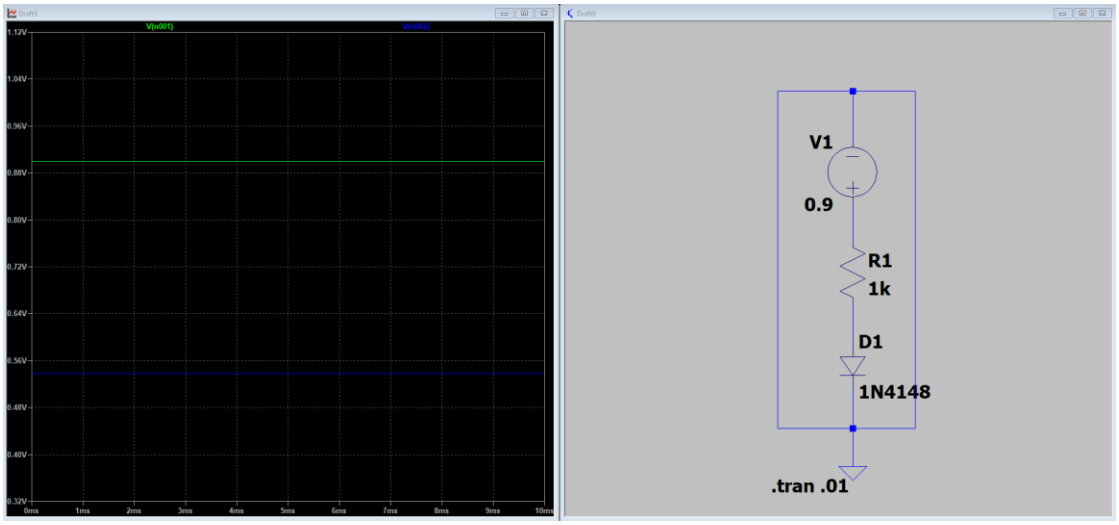
0.6 V


V1
0.7
R1
1k
D1
1N4148
.tran .01

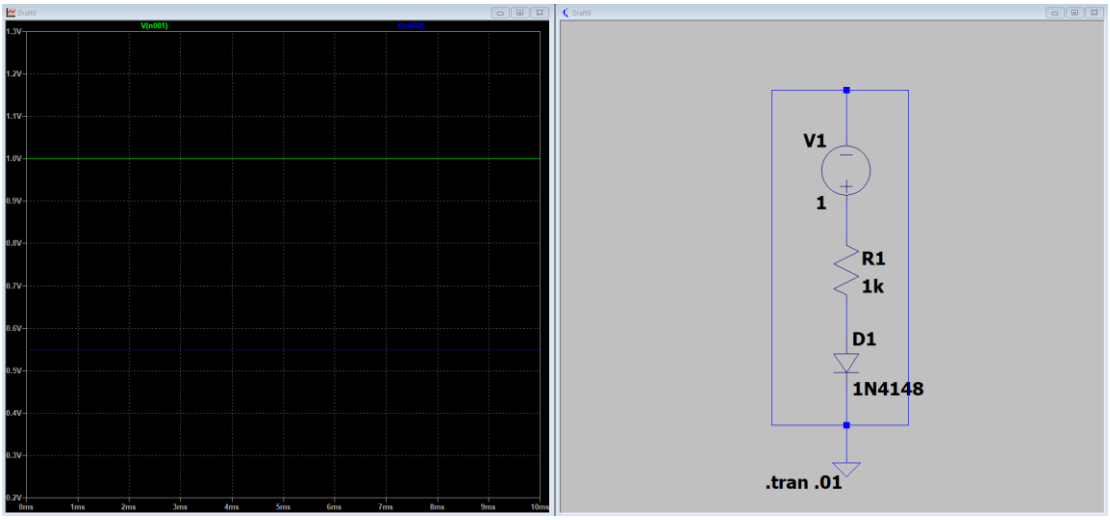
0.7V


V1
0.8
R1
1k
D1
1N4148
.tran .01

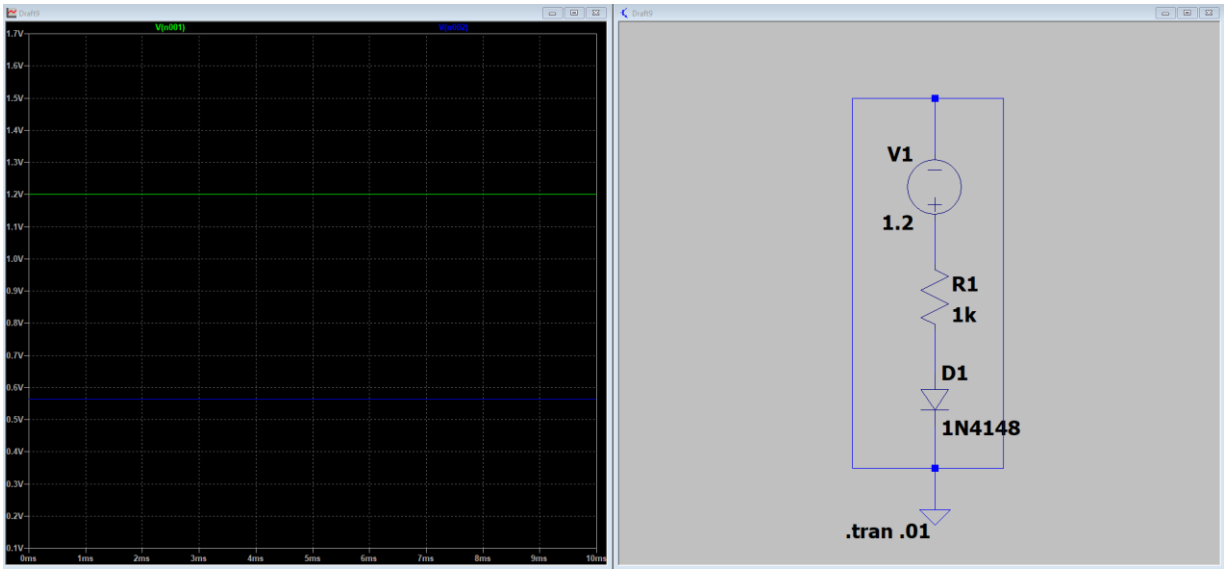
0.8V



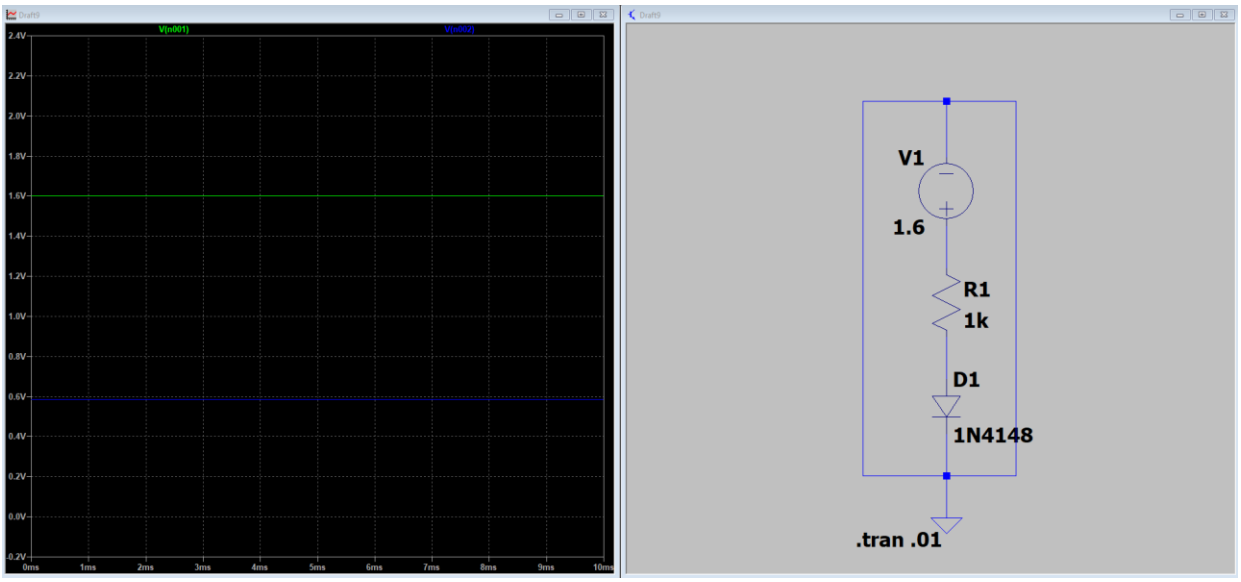
0.9V



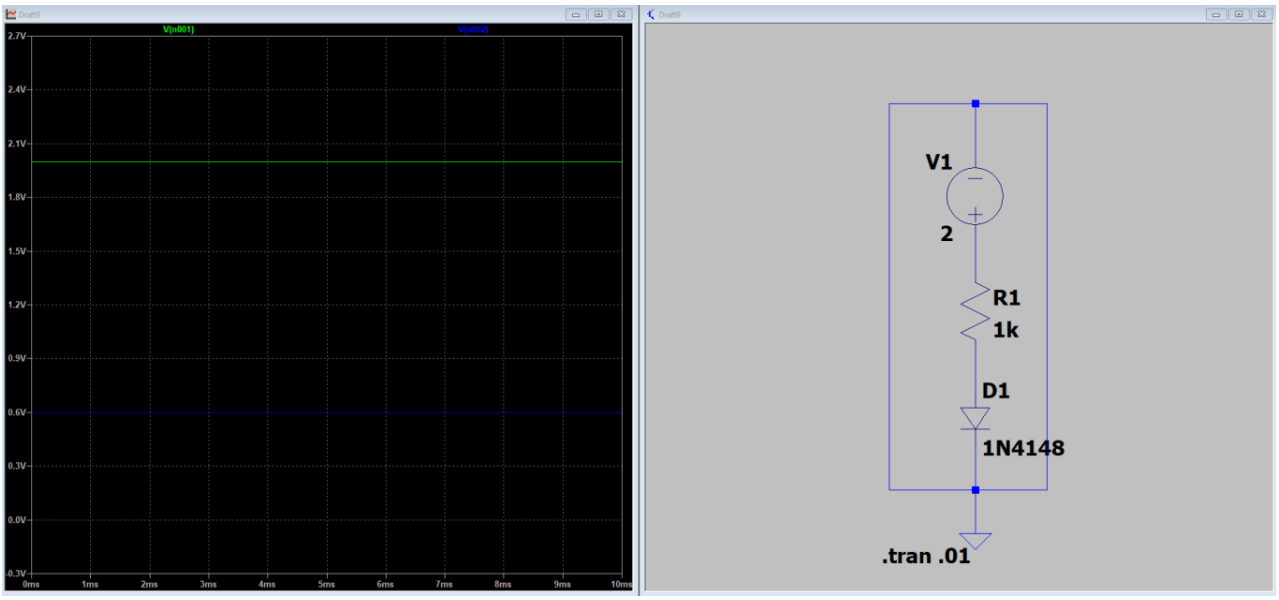
1V


V(n001)
V(n002)
1.7V
1.6V
1.5V
1.4V
1.3V
1.2V
1.1V
1.0V
0.9V
0.8V
0.7V
0.6V
0.5V
0.4V
0.3V
0.2V
0.1V
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
V1
1.2
R1
1k
D1
1N4148
.tran .01

1.2V

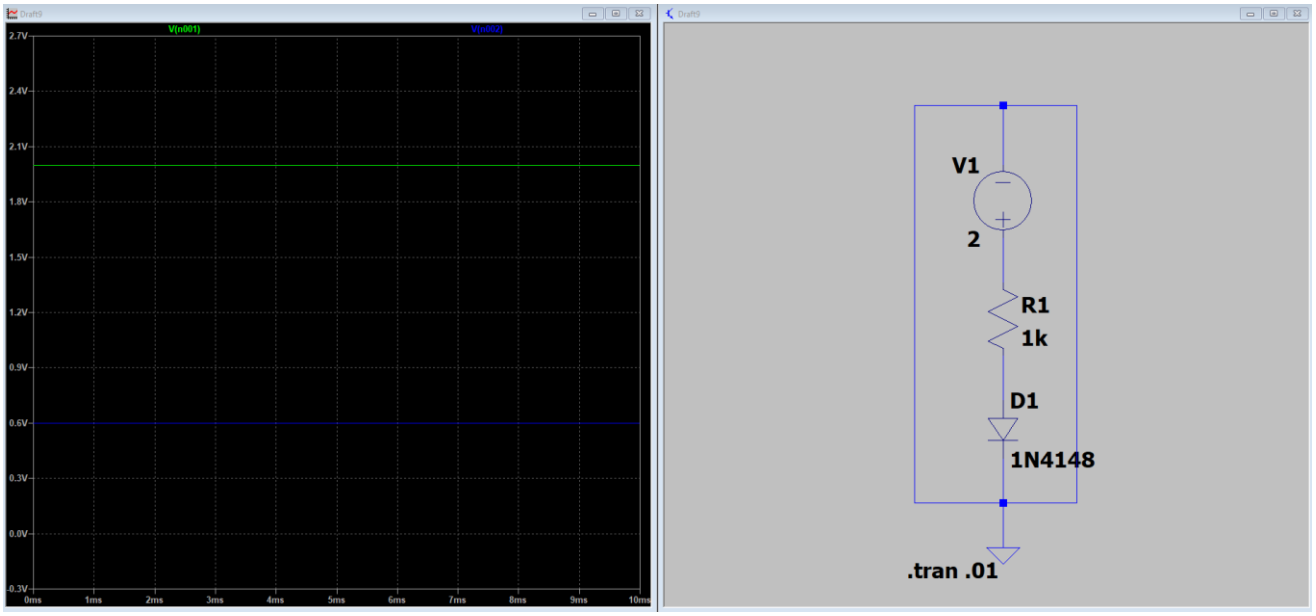

V(n001)
V(n002)
2.4V
2.2V
2.0V
1.8V
1.6V
1.4V
1.2V
1.0V
0.8V
0.6V
0.4V
0.2V
0.0V
-0.2V
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms
V1
1.6
R1
1k
D1
1N4148
.tran .01

1.6V

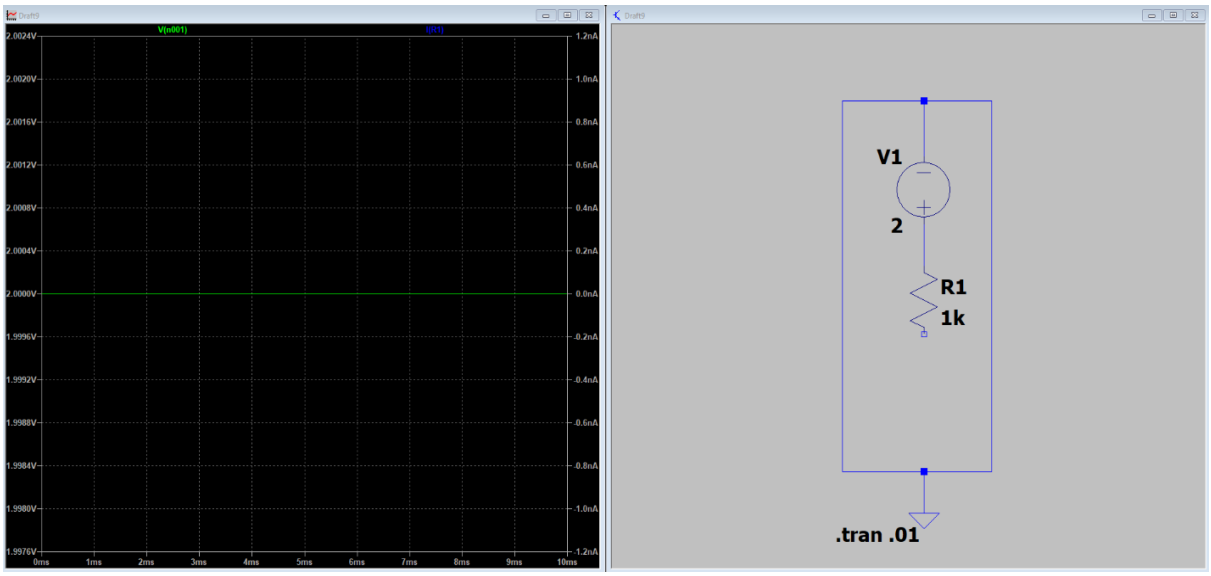


2V

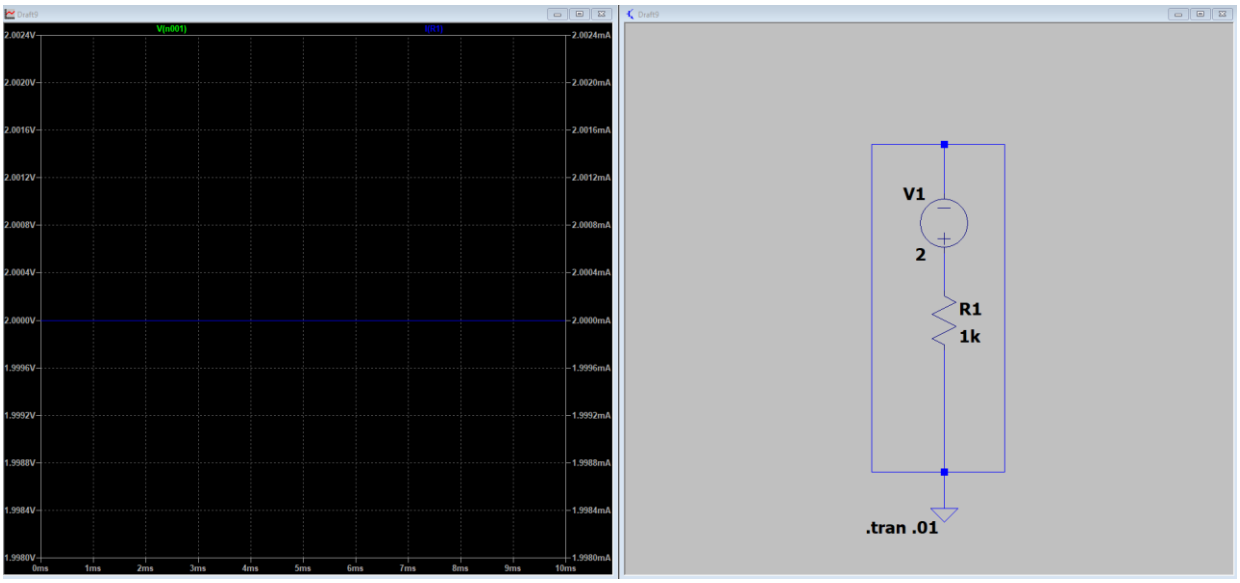
Part 2



Normal Circuit

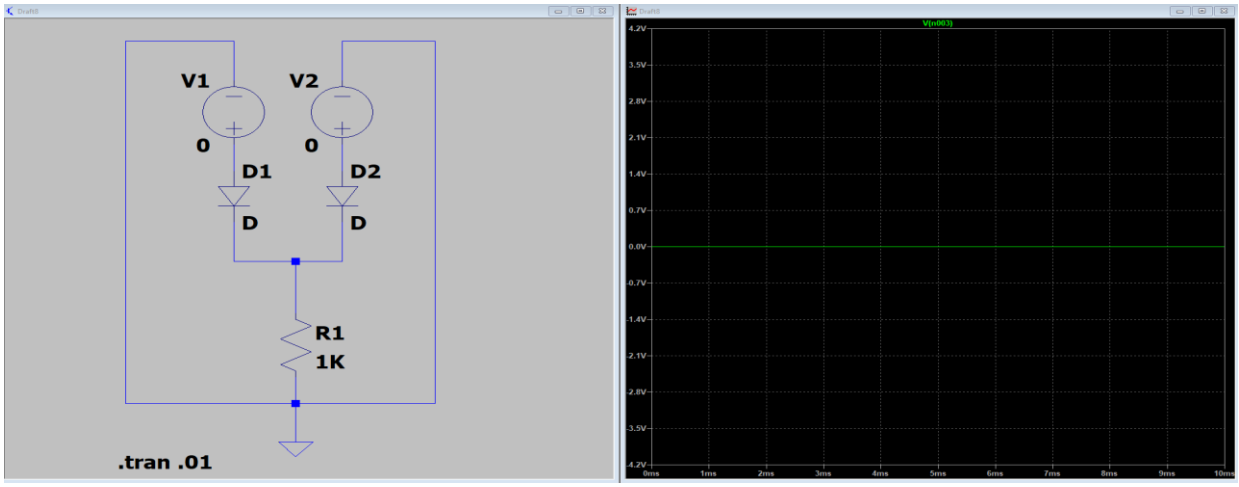


Open Circuit

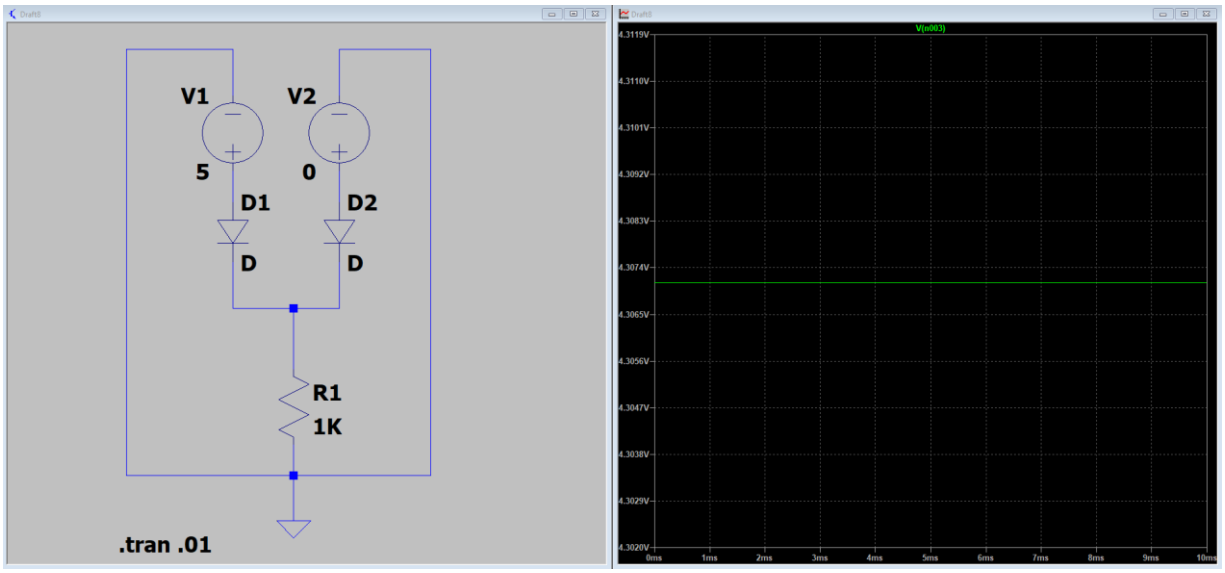


Short Circuit

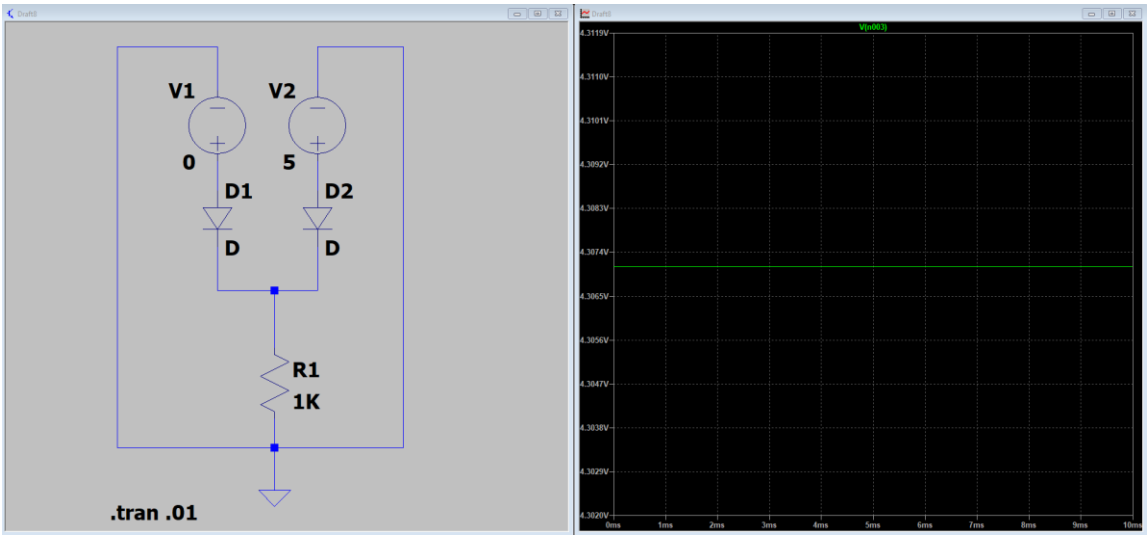
Part 3



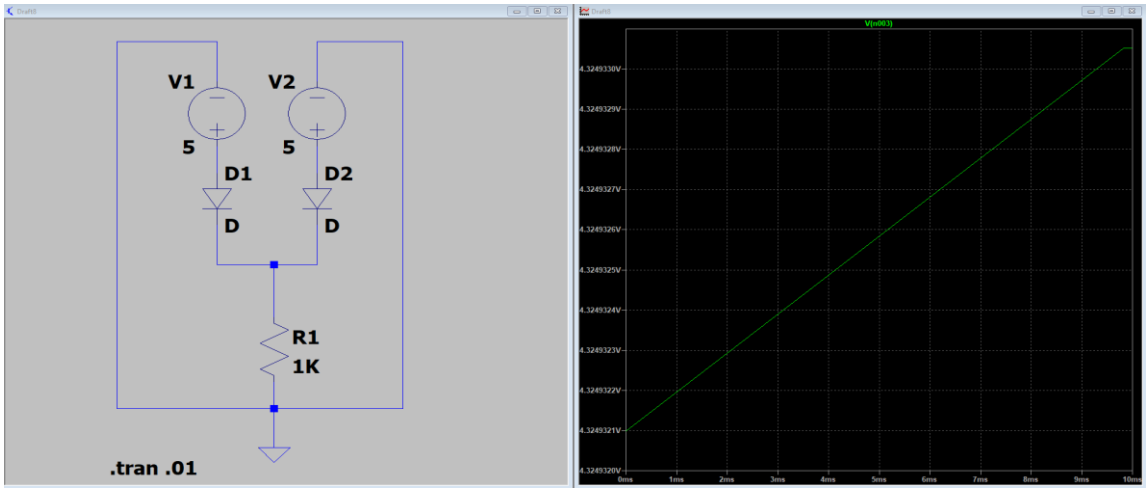
$D_1 = 0V, D_2 = 0V$



$D_1 = 5V, D_2 = 0V$



$D_1 = 0V, D_2 = 5V$



$D_1 = 5V, D_2 = 5V$

FAIRCHILD

November 2014

1N4001 - 1N4007
General-Purpose Rectifiers

Features

- Low Forward Voltage Drop
- High Surge Current Capability



DO-41
COLOR BAND DENOTES CATHODE

Ordering Information

Part Number	Top Mark	Package	Packing Method
1N4001	1N4001	DO-204AL (DO-41)	Tape and Reel
1N4002	1N4002	DO-204AL (DO-41)	Tape and Reel
1N4003	1N4003	DO-204AL (DO-41)	Tape and Reel
1N4004	1N4004	DO-204AL (DO-41)	Tape and Reel
1N4005	1N4005	DO-204AL (DO-41)	Tape and Reel
1N4006	1N4006	DO-204AL (DO-41)	Tape and Reel
1N4007	1N4007	DO-204AL (DO-41)	Tape and Reel

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Value	Unit
V_F	Forward Voltage	I_F = 1.0 A	1.1	V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle	T_A = 75°C	30	µA
I_R	Reverse Current at Rated V_R	T_A = 25°C	5.0	µA
		T_A = 100°C	50	
C_T	Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	15	pF